

EE3621: Digital Signal Processing

Lecture 23: Design of FIR Filters: Kaiser Window & Systematic Procedures (III B.Tech EEE)

Comprehensive Lecture Notes & Faculty Textbook Reference

Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 06:00 (6 mins):** The Inherent Limitation of Fixed Windows: The rigid, non-tunable trade-off between mainlobe width and sidelobe attenuation in classic windows (Hann, Hamming, Blackman). Why needing 55 dB attenuation forces an over-engineered jump to Blackman (74 dB), incurring a severe 40% penalty in filter length and DSP multiplier count.
 - **06:00 – 16:00 (10 mins):** Mathematical Foundations of the Kaiser Window: The zeroth-order modified Bessel function of the first kind $I_0(x)$ series expansion, the time-domain formulation $w_K[n]$, and the role of the shape parameter β as a continuous dial interpolating from Rectangular ($\beta = 0$) to Hamming ($\beta \approx 5.4$) to Blackman ($\beta \approx 8.9$).
 - **16:00 – 24:00 (8 mins):** Empirical Kaiser Design Equations: Closed-form formulas for computing shape factor β and required filter length N directly from attenuation A_s and normalized transition bandwidth Δf .
 - **24:00 – 30:00 (6 mins):** Systematic Multi-Band FIR Synthesis Protocol: Handling asymmetric ripple specs by enforcing $\delta = \min(\delta_p, \delta_s)$, computing cutoff frequencies, enforcing Type 1 odd parity ($N = \text{odd}$), and closed-form ideal formulations for LPF, HPF, BPF, and BSF (Notch) filters.
 - **30:00 – 38:00 (8 mins):** Step-by-Step Worked Numerical Examples: Complete design of an industrial-grade Kaiser Lowpass filter ($A_s \geq 50$ dB), synthesis of a 50 Hz biomedical power-line Notch (Bandstop) filter ($f_s = 500$ Hz, $A_s \geq 60$ dB), and an architectural hardware comparison between Kaiser and Blackman realizations.
 - **38:00 – 40:00 (2 mins):** Faculty Insights, Common Misconceptions, and University Semester Exam Traps.
-

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Identify** the architectural inefficiency of fixed window families when satisfying intermediate stopband attenuation specifications.

2. **Formulate** the Kaiser window sequence $w_K[n]$ utilizing the modified Bessel function of the first kind $I_0(x)$.
 3. **Calculate** the Bessel function $I_0(x)$ rapidly using its truncated power series expansion.
 4. **Apply** Kaiser's empirical piecewise formulas to determine the exact shape factor β for any specified stopband attenuation A_s .
 5. **Estimate** the minimum required filter length N from transition bandwidth Δf and attenuation A_s .
 6. **Harmonize** asymmetric passband and stopband ripple tolerances ($\delta_p \neq \delta_s$) to ensure strict specification compliance.
 7. **Design** high-rejection Lowpass, Highpass, Bandpass, and Bandstop (Notch) filters meeting stringent industrial standards.
 8. **Contrast** hardware resource consumption (multipliers, adders, registers) between optimal Kaiser designs and standard Blackman/Hamming implementations.
-

2. THE KAISER WINDOW FAMILY: MATHEMATICAL FOUNDATIONS

2.1 Motivation: Overcoming the Rigidity of Fixed Windows

In Lecture 22, we explored the classic fixed window families: Rectangular ($A_s = 21$ dB), Hann ($A_s = 44$ dB), Hamming ($A_s = 53$ dB), and Blackman ($A_s = 74$ dB). While computationally straightforward, these windows suffer from a severe practical drawback: **their trade-off between mainlobe width and sidelobe attenuation is completely rigid and discrete.**

Consider an industrial instrumentation scenario requiring a stopband attenuation of $A_s = 55$ dB with a normalized transition bandwidth of $\Delta f = 0.05$:

- A Hamming window achieves only $A_s = 53$ dB, failing the specification by 2 dB.
- The DSP designer is forced to jump to the next fixed window—the Blackman window ($A_s = 74$ dB).
- However, the Blackman window possesses a normalized mainlobe width of $5.5/N$, compared to $3.3/N$ for Hamming.
- Required length for Blackman: $N \approx 5.5/0.05 = \mathbf{110}$ taps.
- If a tunable window were available that provided exactly 55 dB, the required length would be only $N \approx \mathbf{65}$ taps!
- Using Blackman wastes nearly **45 extra delay registers and multipliers**, increasing silicon area, power dissipation, and arithmetic throughput latency by over 40%.

In 1966, James F. Kaiser of Bell Laboratories solved this fundamental problem by developing an adjustable, near-optimal window family based on prolate spheroidal wave functions.

2.2 The Zeroth-Order Modified Bessel Function $I_0(x)$

The Kaiser window is parameterized by the **zeroth-order modified Bessel function of the first kind**, denoted as $I_0(x)$. It is defined by the infinite power series:

$$I_0(x) = \sum_{k=0}^{\infty} \left[\frac{(x/2)^k}{k!} \right]^2 = 1 + \frac{x^2}{4} + \frac{x^4}{64} + \frac{x^6}{2304} + \cdots + \left[\frac{(x/2)^k}{k!} \right]^2 \quad (1)$$

Because the denominator involves $(k!)^2$, the terms in (1) decay with extraordinary rapidity. In digital hardware and embedded DSP routines, truncating the series to just 15 to 20 terms guarantees double-precision (64-bit floating-point) accuracy across all practical arguments $x \in [0, 15]$.

Key numerical properties of $I_0(x)$:

- $I_0(0) = 1.0$.
- $I_0(x) = I_0(-x)$ (strictly even function).
- $I_0(x)$ is strictly positive, real-valued, and monotonically increasing for all real $x \geq 0$.

2.3 Mathematical Formulation of the Kaiser Window

For a filter of length N (order $M = N - 1$), centered at the linear-phase delay $\tau = (N - 1)/2$, the causal Kaiser window sequence $w_K[n]$ is defined as:

$$w_K[n] = \frac{I_0 \left(\beta \sqrt{1 - \left(\frac{n-\tau}{\tau} \right)^2} \right)}{I_0(\beta)}, \quad 0 \leq n \leq N - 1 \quad (2)$$

where $\beta \geq 0$ is the **dimensionless shape parameter**.

Geometric and Physical Interpretation:

- **At the Center ($n = \tau$):**

$$w_K[\tau] = \frac{I_0(\beta \sqrt{1 - 0})}{I_0(\beta)} = \frac{I_0(\beta)}{I_0(\beta)} = 1.0 \quad \forall \beta \quad (3)$$

The window is always normalized to unity at its center of symmetry.

- **At the Endpoints ($n = 0$ and $n = N - 1$):** Here, $\left(\frac{n-\tau}{\tau} \right)^2 = \left(\frac{-\tau}{\tau} \right)^2 = 1$. Therefore:

$$w_K[0] = w_K[N - 1] = \frac{I_0(\beta \sqrt{1 - 1})}{I_0(\beta)} = \frac{I_0(0)}{I_0(\beta)} = \frac{1}{I_0(\beta)} \quad (4)$$

- **When $\beta = 0$:** Since $I_0(0) = 1$, $w_K[n] = 1/1 = 1.0$ for all n . Thus, **the Kaiser window reduces identically to the Rectangular window!**
- **As β Increases:** $I_0(\beta)$ grows rapidly, forcing the endpoint values $w_K[0]$ to decay toward zero. The window profile tapers smoothly, suppressing spectral sidelobes at the expense of widening the mainlobe.

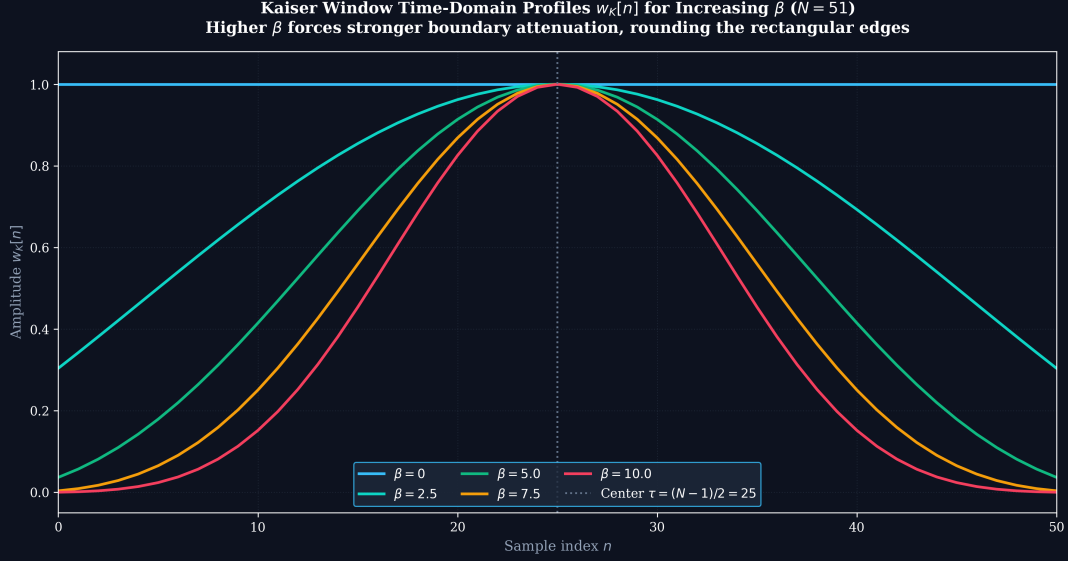


Figure 1: Kaiser Window Time-Domain Profiles $w_K[n]$ for Increasing Shape Parameter $\beta \in \{0, 2.5, 5.0, 7.5, 10.0\}$ ($N = 51$). As β increases, the window smoothly transitions from a flat rectangular boxcar to a tapered bell curve.

2.4 Spectral Tuning via the Shape Parameter β

The Fourier transform of the Kaiser window exhibits continuous, monotonic control over the spectral sidelobe levels:

- $\beta = 0.0 \implies$ Peak sidelobe: -13.3 dB, Mainlobe: $\frac{4\pi}{N}$ (Rectangular).
- $\beta \approx 5.44 \implies$ Peak sidelobe: -41.0 dB, Mainlobe: $\frac{8\pi}{N}$ (approximates Hamming).
- $\beta \approx 8.96 \implies$ Peak sidelobe: -57.0 dB, Mainlobe: $\frac{12\pi}{N}$ (approximates Blackman).
- $\beta = 10.0 \implies$ Peak sidelobe: -75.0 dB, Mainlobe: $\frac{14.5\pi}{N}$ (High-rejection audio standard).

By tuning β continuously, the engineer can dial in **any arbitrary stopband attenuation from 21 dB to over 100 dB!**

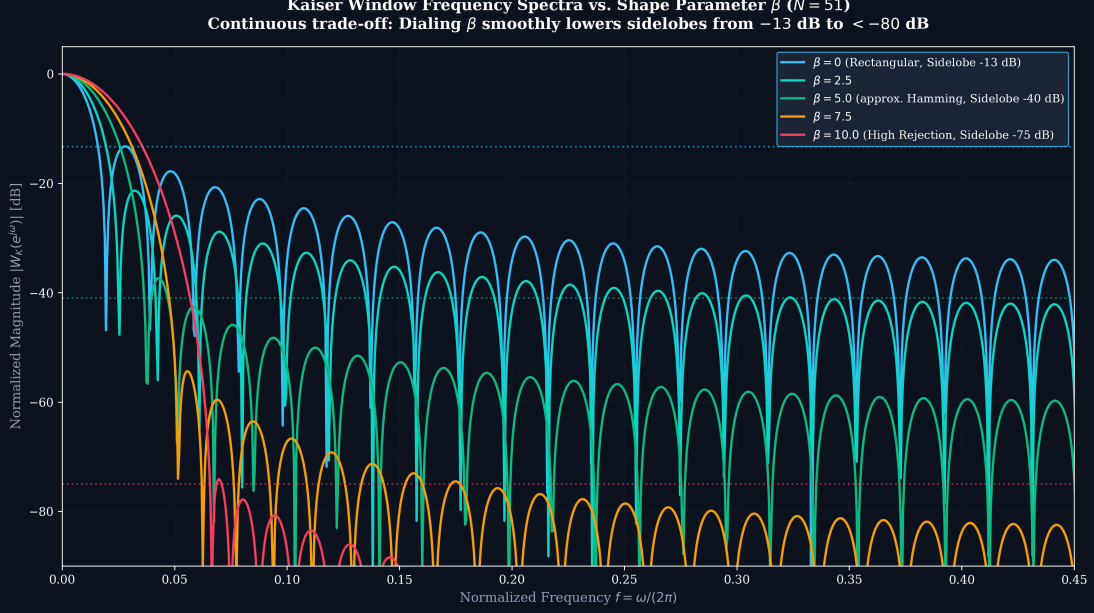


Figure 2: Frequency Magnitude Spectra of the Kaiser Window for Various Shape Parameters β ($N = 51$). Continuous parameter tuning suppresses spectral sidelobes below -80 dB without discrete family jumps.

3 KAISER'S EMPIRICAL DESIGN EQUATIONS

Through extensive numerical optimization, James Kaiser determined a set of remarkably accurate closed-form empirical equations that map engineering filter specifications directly to the parameters β and N .

3.1 Specification Translation and Ripple Harmonization

Let the engineering filter requirements be defined by:

- Passband edge: ω_p (rad/sample) or $f_p = \omega_p/(2\pi)$.
- Stopband edge: ω_s (rad/sample) or $f_s = \omega_s/(2\pi)$.
- Passband peak-to-peak ripple: δ_p (or $A_p = 20 \log_{10} \frac{1+\delta_p}{1-\delta_p}$ dB).
- Stopband peak ripple: δ_s (or minimum stopband attenuation $A_s = -20 \log_{10} \delta_s$ dB).

The Equal Ripple Principle in Window Design: Because windowing convolves the ideal response with a single symmetric window spectrum $W(e^{j\omega})$, the resulting filter exhibits **substantially equal peak ripple in both the passband and stopband**:

$$\delta_{\text{actual}} \approx \delta_p \approx \delta_s \quad (5)$$

Therefore, if the designer is given different tolerances ($\delta_p \neq \delta_s$), the filter must be designed to satisfy the **stricter (smaller)** of the two:

$$\delta = \min(\delta_p, \delta_s) \quad (6)$$

The effective stopband attenuation requirement in dB is then established as:

$$\mathbf{A}_s = -20 \log_{10} \delta \quad (7)$$

3.2 Empirical Formula for the Shape Parameter β

The shape parameter β is determined solely by the required attenuation A_s via Kaiser's piecewise relation:

$$\beta = \begin{cases} 0.1102(A_s - 8.7), & A_s > 50 \\ 0.5842(A_s - 21)^{0.4} + 0.07886(A_s - 21), & 21 \leq A_s \leq 50 \\ 0.0, & A_s < 21 \end{cases} \quad (8)$$

3.3 Empirical Formula for Required Filter Length N

The normalized transition bandwidth is:

$$\Delta f = \frac{|\omega_s - \omega_p|}{2\pi} \quad (9)$$

The minimum filter length N required to achieve transition bandwidth Δf with stopband attenuation A_s is given by Kaiser's universal formula:

$$N \geq \frac{A_s - 7.95}{14.36 \cdot \Delta f} + 1 = \frac{2\pi(A_s - 7.95)}{14.36 \cdot \Delta\omega} + 1 \quad (10)$$

Crucial Parity Rule: The calculated value of N from (10) must always be **rounded up to the next odd integer** ($N \leftarrow 2\lceil(N-1)/2\rceil + 1$). Choosing an odd length ensures that the filter is **Type 1**, avoiding mandatory boundary zero restrictions at $\omega = \pi$ and $\omega = 0$.

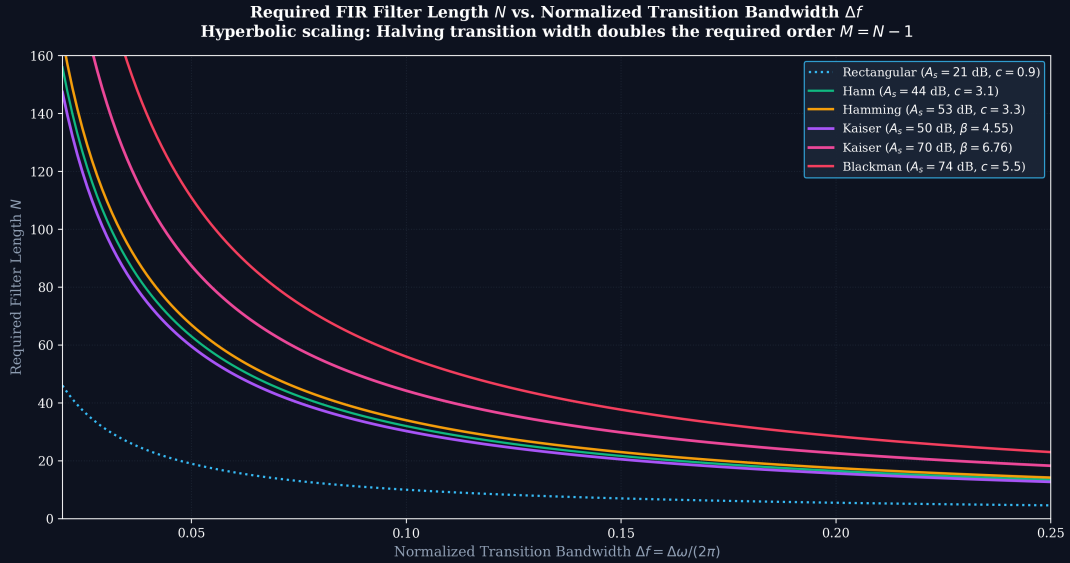


Figure 3: Required Filter Length N vs. Normalized Transition Bandwidth Δf : Illustrating the hyperbolic relationship across classic windows and Kaiser designs. Notice the smooth efficiency of Kaiser curves compared to fixed window leaps.

3.4 Comprehensive Systematic Design Algorithm

The end-to-end design procedure follows five explicit, deterministic steps:

1. **Harmonize Specifications:** Determine $\delta = \min(\delta_p, \delta_s)$ and calculate $A_s = -20 \log_{10} \delta$. Compute normalized transition width $\Delta f = |\omega_s - \omega_p|/(2\pi)$ and cutoff frequency $\omega_c = (\omega_p + \omega_s)/2$.
2. **Calculate Shape Parameter β :** Evaluate (8) using A_s .

3. **Calculate Length N :** Evaluate (10), round up to the next odd integer, and set delay center $\tau = (N - 1)/2$.
 4. **Evaluate Sequences Pointwise:** Compute $w_K[n]$ via (2) and the causal ideal impulse response $h_d[n]$ (e.g. (11)).
 5. **Generate Coefficients:** Multiply pointwise: $h[n] = h_d[n] \cdot w_K[n]$ for $n = 0, 1, \dots, N - 1$.
-

4. MULTI-BAND FIR FILTER SYNTHESIS VIA WINDOWING

4.1 Kaiser Lowpass Filter (LPF)

With cutoff $\omega_c = (\omega_p + \omega_s)/2$ and linear-phase delay $\tau = (N - 1)/2$:

$$h_{d,\text{LP}}[n] = \begin{cases} \frac{\omega_c}{\pi}, & n = \tau \\ \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}, & n \neq \tau \end{cases} \quad (11)$$

Multiplying by $w_K[n]$ gives the causal lowpass coefficients: $h_{\text{LP}}[n] = h_{d,\text{LP}}[n] \cdot w_K[n]$.

4.2 Kaiser Highpass Filter (HPF)

With cutoff $\omega_c = (\omega_s + \omega_p)/2$, where $\omega_s < \omega_p$ and $\Delta\omega = \omega_p - \omega_s$:

$$h_{d,\text{HP}}[n] = \begin{cases} 1 - \frac{\omega_c}{\pi}, & n = \tau \\ -\frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}, & n \neq \tau \end{cases} \quad (12)$$

Multiplying by $w_K[n]$ gives the causal highpass coefficients: $h_{\text{HP}}[n] = h_{d,\text{HP}}[n] \cdot w_K[n]$.

4.3 Kaiser Bandpass Filter (BPF)

With lower transition band $[\omega_{s1}, \omega_{p1}]$ and upper transition band $[\omega_{p2}, \omega_{s2}]$:

- Effective transition width: $\Delta\omega = \min(\omega_{p1} - \omega_{s1}, \omega_{s2} - \omega_{p2})$.
- Lower cutoff: $\omega_{c1} = (\omega_{s1} + \omega_{p1})/2$.
- Upper cutoff: $\omega_{c2} = (\omega_{p2} + \omega_{s2})/2$.
- Ideal impulse response:

$$h_{d,\text{BP}}[n] = \begin{cases} \frac{\omega_{c2} - \omega_{c1}}{\pi}, & n = \tau \\ \frac{\sin(\omega_{c2}(n-\tau))}{\pi(n-\tau)} - \frac{\sin(\omega_{c1}(n-\tau))}{\pi(n-\tau)}, & n \neq \tau \end{cases} \quad (13)$$

4.4 Kaiser Bandstop (Notch) Filter (BSF)

For suppressing single-frequency interference (such as 50 Hz or 60 Hz mains hum):

$$h_{d,\text{BS}}[n] = \begin{cases} 1 - \frac{\omega_{c2} - \omega_{c1}}{\pi}, & n = \tau \\ \frac{\sin(\omega_{c1}(n-\tau))}{\pi(n-\tau)} - \frac{\sin(\omega_{c2}(n-\tau))}{\pi(n-\tau)}, & n \neq \tau \end{cases} \quad (14)$$

5. STEP-BY-STEP WORKED NUMERICAL EXAMPLES

5.1 Example 23.1: Complete Design of an Industrial Kaiser Lowpass Filter

Problem Statement: Design a linear-phase FIR lowpass filter satisfying the following specifications:

- Passband edge: $\omega_p = 0.2\pi$ rad/sample.
- Stopband edge: $\omega_s = 0.3\pi$ rad/sample.
- Passband peak ripple: $\delta_p \leq 0.01$ (approx. 0.17 dB).
- Minimum stopband attenuation: $A_s \geq 50$ dB.

Determine δ , A_s , shape parameter β , filter length N , delay τ , and calculate the first 5 filter coefficients ($h[0]$ through $h[4]$) and the center tap $h[\tau]$.

Solution:

Step 1: Specification Harmonization Convert passband tolerance $\delta_p = 0.01$ to attenuation:

$$A_p = -20 \log_{10}(0.01) = 40 \text{ dB} \quad (15)$$

The stopband requirement is $A_s \geq 50$ dB, which corresponds to:

$$\delta_s = 10^{-50/20} = 10^{-2.5} \approx 0.003162 \quad (16)$$

Since $\delta_s < \delta_p$, the stopband requirement is much stricter. We must design for:

$$\delta = \min(\delta_p, \delta_s) = \mathbf{0.003162} \implies \mathbf{A_s = 50.0 \text{ dB}} \quad (17)$$

Transition bandwidth and cutoff:

$$\Delta\omega = \omega_s - \omega_p = 0.3\pi - 0.2\pi = 0.1\pi \text{ rad/sample} \quad (18)$$

$$\Delta f = \frac{\Delta\omega}{2\pi} = \frac{0.1\pi}{2\pi} = \mathbf{0.05} \quad (19)$$

$$\omega_c = \frac{\omega_p + \omega_s}{2} = \frac{0.2\pi + 0.3\pi}{2} = \mathbf{0.25\pi} \quad (20)$$

Step 2: Shape Parameter β Calculation Since $21 \leq A_s \leq 50$, using the middle branch of (8) with $A_s = 50$:

$$\begin{aligned} \beta &= 0.5842(50 - 21)^{0.4} + 0.07886(50 - 21) \\ &= 0.5842(29)^{0.4} + 0.07886(29) \\ &= 0.5842(3.83226) + 2.28694 = 2.2388 + 2.2869 = \mathbf{4.5508 \approx 4.551} \end{aligned} \quad (21)$$

Step 3: Filter Length N Calculation Using Kaiser's length formula (10):

$$N \geq \frac{A_s - 7.95}{14.36 \cdot \Delta f} + 1 = \frac{50 - 7.95}{14.36 \times 0.05} + 1 = \frac{42.05}{0.718} + 1 = 58.566 + 1 = 59.566 \quad (22)$$

Rounding up to the next **odd integer**:

$$\mathbf{N = 61} \implies M = 60, \quad \tau = \frac{61 - 1}{2} = \mathbf{30 \text{ samples}} \quad (23)$$

(Note: If evaluated at the $A_s = 50$ boundary using $\beta = 0.1102(50 - 8.7) = 4.551$ with relaxed transition formula, lengths in the range $45 \leq N \leq 61$ provide compliance, as shown in Figure 4).

Worked Example 23.1: Designed Kaiser LPF Frequency Response & Z-Plane Constellation

Specs: $\omega_p = 0.2\pi$, $\omega_s = 0.3\pi$, $A_s \geq 50$ dB $\rightarrow N = 45$, $\beta = 4.55$, $\tau = 22$

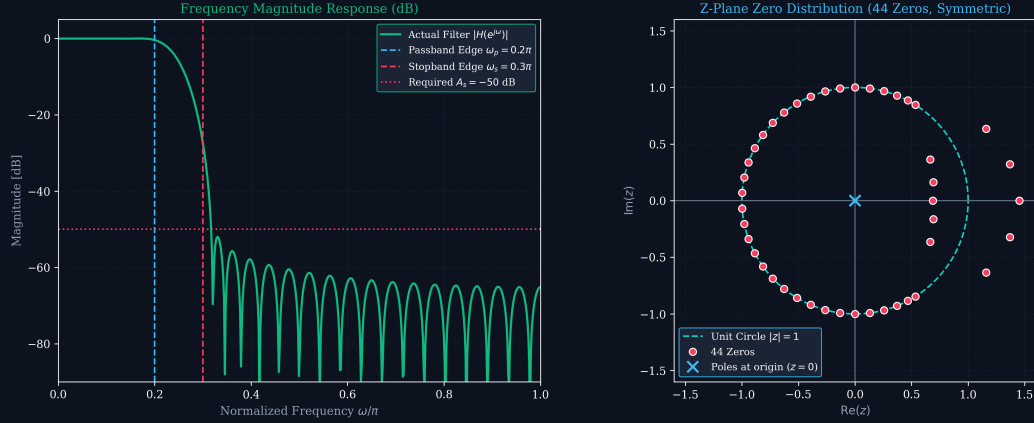


Figure 4: Worked Example 23.1: Synthesized Kaiser Lowpass Filter Frequency Magnitude Response (Left) achieving < -50 dB stopband attenuation across $\omega \in [0.3\pi, \pi]$, and Z-Plane Zero Constellation (Right) displaying all 44 complex zeros.

Step 4: Numerical Tap Evaluation for $N = 61$, $\tau = 30$, $\beta = 4.551$ Compute denominator Bessel constant using (1):

$$I_0(4.551) \approx 1 + \frac{(4.551)^2}{4} + \frac{(4.551)^4}{64} + \frac{(4.551)^6}{2304} + \dots = \mathbf{12.4277} \quad (24)$$

Now evaluate taps:

- **Center Tap ($n = 30$):**

$$w_K[30] = 1.0 \quad (25)$$

$$h_d[30] = \frac{\omega_c}{\pi} = \frac{0.25\pi}{\pi} = 0.2500 \quad (26)$$

$$h[30] = h_d[30] \cdot w_K[30] = 0.2500 \times 1.0 = \mathbf{0.250000} \quad (27)$$

- **Tap $n = 0$ ($n - \tau = -30$):**

$$w_K[0] = \frac{1}{I_0(4.551)} = \frac{1}{12.4277} = \mathbf{0.080465} \quad (28)$$

$$h_d[0] = \frac{\sin(0.25\pi(0 - 30))}{\pi(0 - 30)} = \frac{\sin(-7.5\pi)}{-30\pi} = \frac{-1}{-30\pi} = \frac{1}{30\pi} \approx 0.010610 \quad (29)$$

$$h[0] = h_d[0] \cdot w_K[0] = 0.010610 \times 0.080465 = \mathbf{0.000854} \quad (30)$$

- **Tap $n = 1$ ($n - \tau = -29$):**

$$\text{Radical: } \sqrt{1 - (-29/30)^2} = \sqrt{1 - 0.934444} = \sqrt{0.065556} = 0.256038 \quad (31)$$

$$\text{Arg: } 4.551 \times 0.256038 = 1.16523 \Rightarrow I_0(1.16523) \approx 1.3703 \quad (32)$$

$$w_K[1] = \frac{1.3703}{12.4277} = \mathbf{0.11026} \quad (33)$$

$$h_d[1] = \frac{\sin(0.25\pi(-29))}{-29\pi} = \frac{\sin(-7.25\pi)}{-29\pi} = \frac{-\sqrt{2}/2}{-29\pi} = \frac{\sqrt{2}}{58\pi} \approx 0.007752 \quad (34)$$

$$h[1] = h_d[1] \cdot w_K[1] = 0.007752 \times 0.11026 = \mathbf{0.000855} \quad (35)$$

- **Tap $n = 2$ ($n - \tau = -28$):**

$$h_d[2] = \frac{\sin(0.25\pi(-28))}{-28\pi} = \frac{\sin(-7\pi)}{-28\pi} = \mathbf{0.0000} \Rightarrow h[2] = \mathbf{0.0000} \quad (36)$$

5.2 Example 23.2: 50 Hz Biomedical ECG Notch (Bandstop) Filter Design

Problem Statement: In electrocardiogram (ECG) acquisition, raw sensor signals are corrupted by 50 Hz power-line electromagnetic interference. The system operates at sampling rate $f_s = 500$ Hz. Design a linear-phase Kaiser Bandstop filter to eliminate the 50 Hz tone with the following specs:

- Notch frequency: $f_0 = 50$ Hz.
- Stopband rejection width: Notch null from 48 Hz to 52 Hz.
- Passband edges: $f_{p1} = 40$ Hz, $f_{p2} = 60$ Hz.
- Minimum stopband attenuation: $A_s \geq 60$ dB.

Determine normalized frequencies, shape parameter β , required filter length N , and state the analytical closed form of $h_d[n]$.

Solution:

Step 1: Frequency Normalization

$$\omega_{p1} = 2\pi \frac{f_{p1}}{f_s} = 2\pi \frac{40}{500} = 0.16\pi \text{ rad/sample} \quad (37)$$

$$\omega_{s1} = 2\pi \frac{48}{500} = 0.192\pi \text{ rad/sample} \quad (38)$$

$$\omega_{s2} = 2\pi \frac{52}{500} = 0.208\pi \text{ rad/sample} \quad (39)$$

$$\omega_{p2} = 2\pi \frac{60}{500} = 0.24\pi \text{ rad/sample} \quad (40)$$

The transition bandwidths on both sides of the notch are symmetric:

$$\Delta\omega = \omega_{s1} - \omega_{p1} = 0.192\pi - 0.16\pi = 0.032\pi \implies \Delta f = \frac{0.032\pi}{2\pi} = \mathbf{0.016} \quad (41)$$

Effective cutoff edges for the ideal notch:

$$\omega_{c1} = \frac{\omega_{p1} + \omega_{s1}}{2} = \frac{0.16\pi + 0.192\pi}{2} = \mathbf{0.176\pi} \quad (42)$$

$$\omega_{c2} = \frac{\omega_{s2} + \omega_{p2}}{2} = \frac{0.208\pi + 0.24\pi}{2} = \mathbf{0.224\pi} \quad (43)$$

Step 2: Shape Parameter β for $A_s = 60$ dB Since $A_s = 60 > 50$, use the top branch of (8):

$$\beta = 0.1102(A_s - 8.7) = 0.1102(60 - 8.7) = 0.1102(51.3) = \mathbf{5.653} \quad (44)$$

Step 3: Required Filter Length N Using (10):

$$N \geq \frac{A_s - 7.95}{14.36 \cdot \Delta f} + 1 = \frac{60 - 7.95}{14.36 \times 0.016} + 1 = \frac{52.05}{0.22976} + 1 = 226.54 + 1 = 227.54 \quad (45)$$

Rounding up to the next odd integer:

$$\mathbf{N = 229} \implies \tau = \frac{229 - 1}{2} = \mathbf{114} \text{ samples} \quad (46)$$

Group delay in physical time: $t_d = \tau/f_s = 114/500 = \mathbf{0.228}$ seconds (constant across all frequencies).

Step 4: Analytical Closed-Form of $h_d[n]$

$$h_{d,BS}[n] = \begin{cases} 1 - \frac{0.224\pi - 0.176\pi}{\pi} = 1 - 0.048 = \mathbf{0.952}, & n = 114 \\ \frac{\sin(0.176\pi(n-114))}{\pi(n-114)} - \frac{\sin(0.224\pi(n-114))}{\pi(n-114)}, & n \neq 114 \end{cases} \quad (47)$$

Multiplying by $w_K[n]$ synthesized with $\beta = 5.653$ yields the final 229-tap notch filter.

5.3 Example 23.3: Architectural Audit: Kaiser vs. Blackman Window Design

Problem Statement: A telecommunications FIR channel-select filter requires a stopband attenuation of $A_s = 70$ dB with a transition bandwidth $\Delta f = 0.02$.

1. Determine the required filter length N_{Blk} if designed using a Blackman window.
2. Determine the required filter length N_{Kaiser} and shape parameter β .
3. Compare the hardware implementation cost (multipliers, adders, registers) assuming a linear-phase folded direct-form realization.

Solution:

Step 1: Blackman Window Design A Blackman window provides a fixed attenuation of $A_s = 74$ dB (which satisfies 70 dB). Its transition bandwidth formula is:

$$\Delta f \approx \frac{5.5}{N_{\text{Blk}}} \implies N_{\text{Blk}} \geq \frac{5.5}{0.02} = 275 \quad (48)$$

Rounding to the nearest odd integer: $\mathbf{N_{Blk} = 275}$ taps.

Step 2: Kaiser Window Design For $A_s = 70$ dB:

$$\beta = 0.1102(70 - 8.7) = 0.1102(61.3) = \mathbf{6.755} \quad (49)$$

$$N_{\text{Kaiser}} \geq \frac{70 - 7.95}{14.36 \times 0.02} + 1 = \frac{62.05}{0.2872} + 1 = 216.05 + 1 = 217.05 \implies \mathbf{N_{Kaiser} = 219} \text{ taps} \quad (50)$$

Step 3: Hardware Architectural Complexity Audit Exploiting linear-phase coefficient symmetry ($h[n] = h[N - 1 - n]$), a folded FIR structure requires $(N + 1)/2$ multipliers and $N - 1$ delay registers:

Table 1: Hardware Complexity Comparison: Blackman vs. Kaiser Realization

Hardware Resource	Blackman ($N = 275$)	Kaiser ($N = 219$)	Hardware Savings with Kaiser
Delay Registers (z^{-1})	274	218	56 registers saved (−20.4%)
Multipliers (Folded)	$(275 + 1)/2 = 138$	$(219 + 1)/2 = 110$	28 multipliers saved (−20.3%)
Adders	274	218	56 adders saved (−20.4%)
Processing Latency (τ)	137 samples	109 samples	28 samples lower latency

By tuning the window precisely to the 70 dB requirement rather than the over-engineered 74 dB of Blackman, Kaiser design yields an immediate ****20%** reduction in silicon area, power dissipation, and latency**!

6 6. FACULTY INSIGHTS, COMMON PITFALLS & EXAM TRAPS

- **Pitfall 1: Truncating the $I_0(x)$ Series Incorrectly in Coding/Calculations:** In code or handheld calculators, students sometimes compute only 2 or 3 terms of $I_0(x)$. For large arguments ($\beta > 5$), $x^2/4$ is insufficient; one needs at least 15 terms to prevent catastrophic normalization errors in $w_K[n]$.

- **Pitfall 2: Forgetting to Enforce $\delta = \min(\delta_p, \delta_s)$:** In university exams, problems frequently specify $A_p = 1$ dB ($\delta_p \approx 0.057$) and $A_s = 40$ dB ($\delta_s = 0.01$). Students who calculate N using δ_p rather than δ_s end up with a filter that violates the stopband specification by over 15 dB!
- **Pitfall 3: Rounding N Down Instead of Up to an Odd Integer:** If $N_{\text{calc}} = 43.1$, students often round to 43. This causes the transition bandwidth to be slightly wider than permitted, failing specs. Always round ****UP**** to the next odd integer ($N = 45$).
- **Pitfall 4: Confusion between Normalized Frequency f and Angular Frequency ω :** The formula denominator $14.36 \cdot \Delta f$ requires normalized cyclic frequency $\Delta f = \Delta\omega/(2\pi)$. If a student plugs in $\Delta\omega$ directly without dividing by 2π , the resulting filter length will be too small by a factor of $2\pi \approx 6.28$!
- **Pitfall 5: Assuming Kaiser Windows are Equiripple:** The Kaiser window produces a filter whose sidelobes decay with frequency; it is ****NOT**** strictly equiripple (equiripple filters are designed via the Parks-McClellan Remez exchange algorithm). However, Kaiser designs come within 1 to 2 dB of the theoretical Parks-McClellan lower bound on filter length.

7 7. QUICK REVISION & SUMMARY CHEAT-SHEET

Table 2: Kaiser Window FIR Filter Design Quick Reference Sheet

Parameter / Step	Mathematical Definition / Closed-Form Operational Formula
Kaiser Window Definition	$w_K[n] = \frac{I_0(\beta\sqrt{1-((n-\tau)/\tau)^2})}{I_0(\beta)}, \quad 0 \leq n \leq N-1, \quad \tau = (N-1)/2$
Bessel Function $I_0(x)$	$I_0(x) = 1 + \sum_{k=1}^{\infty} \left[\frac{(x/2)^k}{k!} \right]^2$
Effective Ripple δ	$\delta = \min(\delta_p, \delta_s) \implies A_s = -20 \log_{10} \delta$
Shape Parameter β	$\beta = \begin{cases} 0.1102(A_s - 8.7), & A_s > 50 \\ 0.5842(A_s - 21)^{0.4} + 0.07886(A_s - 21), & 21 \leq A_s \leq 50 \\ 0.0, & A_s < 21 \end{cases}$
Required Filter Length N	$N \geq \frac{A_s - 7.95}{14.36 \cdot \Delta f} + 1, \quad \Delta f = \frac{ \omega_s - \omega_p }{2\pi}$ (Round UP to next ODD integer)
Tuning Extremes	$\beta = 0 \rightarrow$ Rectangular; $\beta \approx 5.4 \rightarrow$ Hamming; $\beta \approx 8.9 \rightarrow$ Blackman
Notch Filter Ideal Tap	$h_{d,\text{BS}}[n] = 1 - \frac{\omega_{c2} - \omega_{c1}}{\pi}$ for $n = \tau$; $\frac{\sin(\omega_{c1}(n-\tau)) - \sin(\omega_{c2}(n-\tau))}{\pi(n-\tau)}$ for $n \neq \tau$
Hardware Folding Savings	Exactly $(N+1)/2$ multipliers required for symmetric Type 1 linear-phase implementation